

Review

Knowledge graph-based manufacturing process planning: A state-of-the-art review

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ABSTRACT

Computer-aided process planning is the bridge between computer-aided design and computer-aided manufacturing. With the advent of the intelligent manufacturing era, process knowledge is important for process planning. Knowledge graph is a semantic representation method of knowledge that has attracted extensive attention from the industry and academia. Process planning using the process knowledge graph has become an important development direction for computer-aided process planning. From the analysis of the published reviews, there have been many computer-aided process planning reviews with different focuses. We focus on the techniques and applications of knowledge graph in manufacturing process planning. Therefore, this paper comprehensively reviews knowledge graphs in manufacturing process planning. We analyze the key technologies of process knowledge graph, including process knowledge representation, process knowledge extraction, process knowledge graph construction, process knowledge graph refinement, process knowledge graph validation, and process generation. We also explore the combination of process knowledge graphs and large language models. Finally, potential future research directions are proposed.

1. Introduction

Process planning is the intermediate stage between product design and product production, which defines processing methods, processing sequences, process parameters, machines, and tools for production. Therefore, product quality, manufacturing efficiency, and cost are related to process planning. Depending on the product, process planning presents various manifestations. For example, machining process planning includes path planning, tool and device selection, and time calculation. In traditional process planning, process designers use existing knowledge and experience to design the process manually. Since the 1960s, computers have replaced manual methods for process planning, forming computer-aided process planning (CAPP) techniques. In contrast to manual design, CAPP can significantly improve the ability to handle complex process planning. In 1965, Niebel first proposed the CAPP [1]. After over half a century of development, the CAPP has been a key technology in computer-integrated manufacturing (CIM). As a bridge between computer-aided design (CAD) and computer-aided manufacturing (CAM), the CAPP converts product designs into optimal manufacturing sequences and allocates rational manufacturing resources. The presence of CAPP has increased the automation and computerization of process planning. In addition, it reduces the

difficulty for process designers to manage process data and enhances process design efficiency.

In recent decades, artificial intelligence (AI) has experienced rapid development [2]. The use of AI to resolve industrial manufacturing issues is a hot topic for research. AI can simulate human behaviors such as perception, learning, inference, and decision-making, which can be divided into perceptual intelligence and cognitive intelligence. Knowledge graph (KG) is an important research direction of cognitive intelligence that originates from the traditional Knowledge-Based Expert System (KB-ES) [3]. KG is a structured representation of knowledge, which is widely used in intelligent inference, intelligent decision-making, semantic search, intelligent question and answer (Q&A), recommender systems, semantic understanding, visual understanding, big data analytics, and device interconnection [4]. The traditional KB-ES is widely used in process planning, proving that KG can be used in process planning. KG is more intelligent than traditional KB-ES, and many researchers have used the process knowledge graph (PKG) for process planning in the past several years.

The development of AI technology has given new momentum to the development of the manufacturing industry and fostered the transformation of the manufacturing industry from automation to intelligence.

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This requires process planning to develop in the direction of intelligence to satisfy the requirements of smart manufacturing. How to accurately represent and utilize process knowledge in intelligent process planning is important. This process knowledge is obtained from expert experiences, process documents, and design documents. As a result, there is a lack of effective organization and management of process knowledge and an inability to develop an effective and accurate knowledge base. As a branch of AI, KG is an effective tool for knowledge management [5]. Process knowledge extraction, PKG construction, and process route generation are important for intelligent process planning. At present, many researchers have applied KG to process planning, and there has become a certain tendency to expand the field of manufacturing using KG techniques. As well, many relevant scientific results have been obtained. The existing research results show that PKG-based process planning has the following advantages compared to traditional CAPP: (1) PKG can more easily store and handle complex process knowledge of multiple modalities, including structured process knowledge, semi-structured process knowledge, and structured process knowledge [6–8]; (2) Using the PKG can reveal the explicit and implicit relations between process knowledge, and can make the seemingly isolated process knowledge interconnected [9]. The structure of the PKG determines the ability to complete the process knowledge; (3) Process generation based on PKG is the globally optimal result by combining all process knowledge of the graph [10]. Compared with the traditional CAPP, it is more generalized and smarter [11]; (4) The PKG-based process planning can be applied to large-scale dynamic manufacturing processes due to its flexibility [12]. However, traditional CAPPs are not suitable for large-scale dynamic manufacturing processes because they cannot adjust the process and balance the resource load in real time [13]; (5) PKG is a key technology for transforming traditional manufacturing into intelligent manufacturing [14,15]. Therefore, we review the knowledge graph-based process planning techniques and analyze the future development trends in this paper. With the analysis of published reviews, this paper differs from previous works. We focus on the key technologies of knowledge graph-based process planning. We hope this review will be able to contribute to process planning research and provide reference material for implementation.

In conclusion, the main contributions of our review paper are as follows:

(1) From the published reviews, it can be observed that numerous works have been done to review CAPP systems based on non-knowledge graph approaches. These works mainly focused on the review of more mature methods, such as feature-based, expert system-based, etc. Since 2012, no comprehensive review work focused on knowledge graph-based manufacturing process planning.

(2) We focus on knowledge graph-based manufacturing process planning, which is the newest technique in the CAPP field. We think this is one of the future directions for manufacturing process planning, so the relevant research results of the previous decade should be reviewed and analyzed.

(3) We comprehensively review knowledge graph-based manufacturing process planning from various perspectives, including key technologies, related applications, and future developments.

(4) We also summarize the development of computer-aided process planning and traditional non-knowledge graph-based methods. This allows comparison with the knowledge graph-based methods.

The rest of this paper is organized as follows. Section 2 reviews the related review works. Section 3 provides basic concepts for process planning. Section 4 describes the knowledge graph. Section 5 illustrates the knowledge graph for process planning. Section 6 explores the combination of process knowledge graphs and large language models. Section 7 summarizes the current applications of PKGs. Section 8 discusses possible future directions. Section 9 concludes this paper.

2. Related review works

2.1. Previous reviews

We retrieved published CAPP review articles for a more detailed insight into the evolution of CAPP systems. There have been 23 review articles since 1982. Through comparative analysis, we summarized the publication date, number of references and citations, and technical key focus for each review, as shown in Table 1. The published reviews are summarized below in a paragraph for each decade.

In 1982, Weill et al. reviewed the application of computers in process planning and classified systems according to the characteristics of systems [16]. In 1984, Harold discussed decision logic to build manufacturing methods and strategies that help design generative systems [17]. In 1985, Eversheim et al. reviewed CAPP systems and predicted future directions [18]. In 1988, Ham et al. analyzed the current state of research in CAPP and illustrated the potential of artificial intelligence for applications in logically extended integrated planning frameworks [19]. In 1989, Gupta et al. summarized the application of expert systems in process planning and manufacturing [20]. In the also year, Alting et al. summarized more than 200 papers and 14 well-known CAPP systems. The future trends suggested by the authors include interfaces, integration, artificial intelligence technologies, and individual database systems [21].

In 1993, Eversheim et al. reviewed CAPP in the industrial environment, classified CAPP systems, and indicated the trends in CAPP software [22]. In 1995, Kirtsis reviewed 52 process planning systems based on knowledge expert systems and summarized the techniques for combining process planning and knowledge expert systems [23]. Michael et al. reviewed CAPP based on feature recognition techniques and the use of CAD/CAM [24]. With the rapid development of hierarchical manufacturing. In 1997, Marsan et al. reviewed process planning techniques in hierarchical manufacturing, which summarized part positioning, support structure generation, path selection, and process parameters [25]. In 1998, Marri et al. reviewed CAPP literature (1989–1996) based on system types, presented the advantages and disadvantages of various systems, and proposed future research directions [26]. Zha et al. reviewed process planning methods for the assembly domain. The authors proposed an integrated intelligent method for assembly-oriented design and planning based on parallel engineering and artificial intelligence [27].

In 2000, Kulkarni et al. reviewed the process planning techniques in layered manufacturing again and predicted the future directions [28]. In 2001, Ahmad et al. reviewed the development of CAPP research, including feature-based design, object-oriented programming, technical databases, expert systems, and artificial intelligence [29]. In 2007, Zhang et al. analyzed how agent technology can support collaborative process planning and summarized collaborative process planning systems based on agent technology [30].

In 2010, Xu et al. reviewed CAPP papers of the past decade and summarized feature-based techniques, knowledge-based systems, artificial neural networks, genetic algorithms, fuzzy set theory and fuzzy logic, Petri nets, agent-based methods, Internet-based techniques, STEP-compliant CAPP, and other emerging technologies [31]. In 2014, Yusof et al. reviewed the CAPP literature for the past 12 years (2002–2013) and represented the review graphically to understand the past, present, and future of CAPP [32]. In 2017, Kumar reviewed the application of artificial intelligence in CAPP and manufacturing. The research work of the last 35 years (1981–2016) is analyzed in terms of feature-based design, expert systems, evolutionary methods, scheduling, manufacturing, and other applications [33]. In 2018, Li et al. reviewed the application of knowledge representation methods in machining process planning for machining. A total of 698 papers related to knowledge representation methods are discussed, introducing feature identification, tool selection, mechanism planning, operation selection and sequencing, and computer numerical control (CNC) machining plan generation [34].

Table 1
Comparison of published reviews about CAPP.

No.	Author	Year	Title	Ref	Cited	Key focus
1	Weill et al. [16]	1982	Survey of Computer Aided Process Planning Systems	28	147	Computer usage and issues
2	Steudel [17]	1984	Computer-aided process planning: past, present and future	18	117	Generative systems
3	Eversheim et al. [18]	1985	Survey of Computer Aided Process Planning Systems	\	80	Process planning
4	Ham et al. [19]	1988	Computer-aided process planning: the present and the future	60	298	CAPP systems
5	Gupta et al. [20]	1989	A survey of expert systems in manufacturing and process planning	33	85	Expert systems
6	Alting et al. [21]	1989	Computer aided process planning: the state-of-the-art survey	206	684	CAPP systems
7	Eversheim et al. [22]	1993	Computer-aided process planning-State of the art and future development	17	88	CAPP systems and software
8	Kiritsis [23]	1995	A review of knowledge-based expert systems for process planning. Methods and problems	133	167	Knowledge-based expert systems
9	Michael et al. [24]	1995	An overview of automatic feature recognition techniques for computer-aided process planning	44	170	Automatic feature recognition techniques
10	Marsan et al. [25]	1997	Survey of Process Planning Techniques for Layered Manufacturing	75	58	Layered manufacturing
11	Marri et al. [26]	1998	Computer-aided process planning: A state of art	34	227	CAPP systems and issues
12	Zha et al. [27]	1998	Integrated intelligent design and assembly planning: A survey	205	95	Intelligent design
13	Kulkarni et al. [28]	2000	A review of process planning techniques in layered manufacturing	84	541	Layered manufacturing
14	Ahmad et al. [29]	2001	CURRENT TRENDS IN COMPUTER AIDED PROCESS PLANNING	42	79	AI techniques
15	Zhang et al. [30]	2007	Agent technology for collaborative process planning: a review	55	102	Agent technologies
16	Xu et al. [31]	2011	Computer-aided process planning-A critical review of recent developments and future trends	215	405	Classification and trends
17	Yusof et al. [32]	2014	Survey on computer-aided process planning	121	174	Classification and trends
18	Kumar et al. [33]	2017	State of The Art-Intense Review on Artificial Intelligence Systems Application in Process Planning and Manufacturing	368	136	AI systems and applications
19	Li et al. [34]	2018	A survey of knowledge representation methods and applications in machining process planning	152	34	Knowledge representation methods
20	Wswasi et al. [35]	2018	A survey on smart automated computer-aided process planning (ACAPP) techniques	150	83	Automatic CAPP
21	Kumar [36]	2018	Knowledge-based expert system in manufacturing planning: state-of-the-art review	154	92	Knowledge-Based Expert Systems
22	Besharati et al. [37]	2019	Intelligent process planning for smart manufacturing systems: a state-of-the-art review	37	15	Intelligent process planning
23	Wen et al. [38]	2022	The key technologies of machining process design: a review	145	\	3D process design
	Our	2023	Knowledge Graph-based Manufacturing Process Planning: A State-of-the-art Review	233	\	Manufacturing PKGs

Table 2
Comparison of various typical CAPP systems.

Type	Principles	Features
Index	The Index CAPP systems are simple and easy to set up but have low automatic decision-making capability. Suitable for mass production with high part process similarity.	Simple and easy to set up, but with low automatic decision-making capability. Suitable for mass production with strong part process similarity.
Variant	The Variant CAPP systems are based on grouping technology, which classifies parts into partial clusters according to the similarity of part structure and process. Each part cluster is formulated with corresponding processes, which can be retrieved based on the code.	The system structures are simple, mature, and stable. However, the systems are inflexible and can only be developed for specific products.
Generative	The Generative CAPP systems automatically generate processes through logical inferences based on the input information.	Processes can be automated for decision-making. However, most processes are difficult to build appropriate algorithms for and can only handle processes in specific environments.
Synthesis	The Synthesis CAPP systems combine Variant and Generative. The systems first retrieve the standard process for the part cluster and then modify the standard process using automatic decision-making.	Avoid the limitations of Variant systems and the difficulty of Generative systems.

A review of intelligent automated computer-aided process planning techniques is reviewed by Wswasi et al. The review investigated the automatization of computer-aided process planning systems and automatic feature recognition techniques [35]. Kumar et al. reviewed knowledge-based expert systems for process planning. The authors summarized the applications of expert systems in product diversity, process planning activities, machining, tool selection, tool design, welding, advanced manufacturing, and product development between 1981 and 2016 [33]. In 2019, Besharati et al. reviewed intelligent process planning in smart manufacturing systems. The article described feature representation and identification techniques, knowledge-based expert systems, artificial intelligence techniques, CAD/CAPP system integration, and decision automation [37]. In 2022, Wen et al. reviewed key technologies of machining process design, including the current status and trends in 3D process planning technology. This work analyzed process information management, work process modeling, and the application of 3D processes on the shop floor [38].

2.2. Review gaps

The published reviews show that the existing works can be divided into the following categories: feature-based recognition [24], knowledge-based representation [34], expert-based system [20,23,36], intelligent system-based [33,37], etc. It can be observed that few review works for KG-based manufacturing process planning have been conducted since 2012, which is also because the application of knowledge graphs in the manufacturing industry is still developing. In contrast to previous works, this paper focuses on KG-based manufacturing process planning. KG plays a significant role in intelligent process planning and will be the trend for the future. PKG can improve the efficiency of process planning and shorten the cycle. In this paper, we review the key technologies of KG in process planning and their applications. We hope to provide references for researchers in this field.

2.3. Proposed methodology

Because KGs and process planning are the fundamental theories of this review, we first describe the basic theories of these two. Compared to the half-century development history of process planning, the KG was formally introduced by Google in 2012. Therefore, the development history of knowledge graph-based process planning is relatively short. For these reasons, we review knowledge graph-based process planning in the form of key techniques. The representation of process knowledge, extraction of process knowledge, construction of process knowledge graph schema layer and data layer, and knowledge graph-based process generation are reviewed. Lastly, potential future directions are discussed.

3. Brief overview of process planning

Process planning is the headwind of manufacturing, which is the transition between design and manufacturing. Process planning defines parameters, resources, routes, tools, machines, and manufacturing time [39]. Since Niebel proposed computers as an alternative to manual process planning, CAPP has become dominant [1]. Traditional CAPP system is classified into several types such as index, variant, generative, and synthesis. The features of these systems are summarized in Table 2. Among them, generative and variant are the most widespread in manufacturing, and their structured workflows are shown in Fig. 1. From a technical standpoint, several technologies are applied to the CAPP system. For example, feature techniques, knowledge expert systems, neural networks, genetic algorithms, fuzzy theory, agent techniques, Petri nets, etc.

Since the first CAPP system was born in 1976, the technologies mentioned above have been gradually applied [32]. Feature technology includes feature recognition and feature design. Many processes use features as input, representing the geometry data of the part in implementing CAPP. Sormaz et al. proposed a scheme for identifying geometric features of parts and knowledge representation that associates features with process knowledge. This demonstrates that KB-ES can more easily infer process planning [40]. KB-ES is a research direction of AI, including databases, knowledge base, and inference mechanism [33]. KB-ES obtains knowledge from process experts and simulates the problem-solving skills of process experts [36]. In contrast to KB-ES, neural networks can recognize complex features without the need to input prior knowledge into the system. Neural networks are trained to have the ability to infer rules or knowledge. Devireddy et al. proposed to adopt neural networks for process planning, and the trained system can handle new components [41]. Ming et al. combined KB-ES with neural networks to form a hybrid intelligent inference model for CAPP, which enhanced the suitability of the system for dynamic manufacturing environments [42]. The genetic algorithm is an intelligent algorithm that has been used for global search and optimization in scientific engineering. Applying genetic algorithms for CAPP can find the globally optimal process planning. For example, Mak et al. used genetic algorithms to optimize Hopfield neural networks, which can optimize the process scheme [43].

In the actual process, the targets and constraints are often ambiguous. Chang integrated expert systems, neural networks, and fuzzy theory into the CAPP system [44]. The integrated CAPP system enhanced the ability to deal with dynamic and uncertain problems. Many researchers have combined Petri nets with process planning to improve the flexibility of process planning and meet the requirements

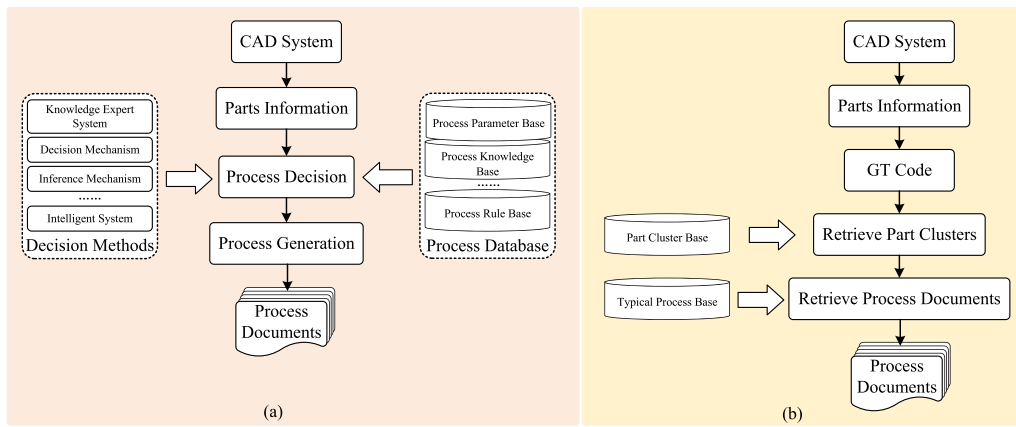


Fig. 1. The basic CAPP workflow, (a) denotes a generative CAPP system and (b) is a variant CAPP system.

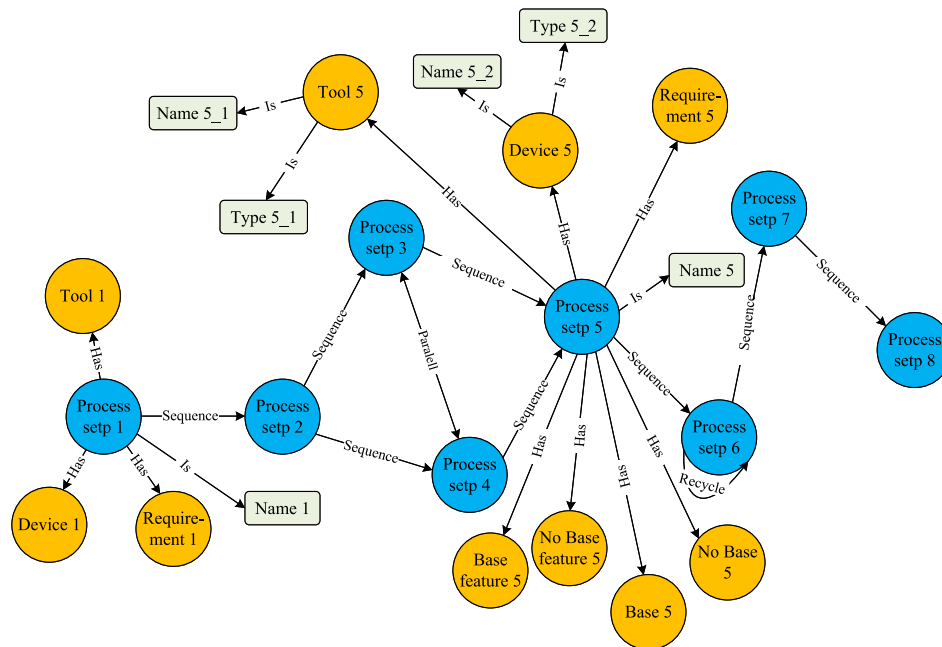


Fig. 2. Structure of a simple PKG. The blue nodes represent the process steps, the orange nodes represent the resources required for the process steps, and the dark green nodes represent the attributes. The relations between process steps include sequence, parallel, and recycle. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

of dynamic planning. The use of agent technology realized the large-scale distributed CAPP system. Complex process planning tasks were assigned to different specialized systems, thus building distributed, parallel, and collaborative systems [45,46].

4. Brief overview of knowledge graph

KG is a structured form of knowledge representation, which uses graph structures to model and store knowledge. The KG was proposed by Google in 2012 [47]. Google has used the KG on the web search engine to achieve recognition of semantic knowledge in search and link search results. Since the KG was formally introduced, it has been rapidly developed in recent years as an important part of AI [48]. It has been applied in natural language processing (NLP), intelligent Q&A systems, intelligent inference systems, intelligent recommender systems, etc. KG can also be regarded as an intelligent system integrating knowledge and data. In recent years, some large-scale KGs have been published, including DBpedia [49], YAGO [50], Wikidata [51], Freebase [52], SUMO [53], NELL [54], and BabelNet [55]. Statistics of some KGs are shown in Table 3.

KG is a heterogeneous graph that contains multiple types of nodes and relational edges. Fig. 2 shows a simple PKG. KGs are represented in three forms, which are property graph (PR) [56], resource description framework (RDF) [57], and ontology web language (OWL) [58]. The PR is a directed graph that consists of nodes, edges, and attributes. Although it is a flexible approach, it lacks deep semantic information. The basic component of the RDF is the relational triple (head entity, relation, tail entity), and multiple relational triples make up the knowledge graph [59]. In contrast to the PR, the RDF contains semantic information. OWL is an RDF extension that adds more semantic expression components [60]. The OWL has powerful representation capabilities to obtain multiple attributes of entities through different edges and to discover high-level relations of entities through relational links.

As KG has shown excellent application potential, it has attracted considerable attention from academia and industry. The combination of KG and process planning is one of the important foundations for achieving intelligent manufacturing. PKG is the product of the combination. Hence, KG in process planning is the highlight of the following section.

Table 3
Statistics of some generic KGs.

KG	Entities	Facts	Types	Relations
DBpedia [49]	5044233	400	760	1355
YAGO [50]	6349359	479392870	819292	77
Wikidata [51]	52252549	732420508	2356259	6236
Freebase [52]	49947845	3041722635	26507	37781
NELL [54]	5120688	60594443	1187	440
BabelNet [55]	7735436	178982397	6044564	22

5. Knowledge graph in process planning

PKG is a domain-specific KG. PKGs mainly manage process knowledge and perform manufacturing process planning [61]. Generic KG is a structured knowledge base for generic domains, containing a lot of commonsense knowledge from the open world. Numerous generic KGs are published [62], as shown in Table 3. Therefore, the main difference between PKGs and generic KG are data sources and object-oriented. Process knowledge are dynamic and highly contextual, as well as describe process entities that change over time. Open-world knowledge are robust and may not change for a long time once established [63]. The PKGs are also similar to the generic KG in many aspects. First, many methods used in generic KGs can be migrated to PKGs, such as knowledge extraction, entity alignment, knowledge fusion, etc. Second, some mature tools can be used for PKGs, such as the Neo4j database. Therefore, the next sections focus closely on PKGs, while introducing some techniques for PKGs.

Process knowledge representation (PKR) is the description of process knowledge and is the foundation of the data structure for constructing PKG. Thus, the PKRs are summarized. The construction of PKG is the central task of the knowledge graph in process planning. Therefore, the key procedures and techniques in the construction process are reviewed, including process knowledge extraction, schema layer construction, and data layer construction. Finally, the methods of process generation on the PKG are introduced.

5.1. Process knowledge representation

Process knowledge includes rules, standards, and experiences in the process planning procedure. PKR is the methodology, framework, and structure to organize and store process knowledge data. Researchers have done many studies on PKR methods. Fig. 3 shows the milestone techniques for knowledge representation development, and Table 4 compares the features of these techniques. Several key PKR approaches are summarized below.

Ontology-based knowledge representation is a hierarchical representation method that can represent rich semantic features and ensure semantic consistency. Zhang et al. proposed an ontology-based process knowledge representation for knowledge of resources, tool selection, and machining strategies in machining operations [64]. García et al. represented and identified the features of sheet metal parts based on an ontology model [65]. Liu et al. used an ontology-based knowledge modeling approach for modeling CAD and generic information in additive manufacturing process planning [66]. Eum et al. proposed an ontology-based process planning knowledge modeling approach to represent the process planning knowledge for selecting machining operations for multi-axis machining features [67]. Huang et al. integrated assembly requirements, spatial information, assembly operations, and assembly resources into an assembly process design knowledge ontology [68]. Solano et al. used the MIRC (manufacturing and inspection resource capability) ontology to represent the activities and capabilities of manufacturing resources [69–71]. Kang et al. combined an ontology-based representation model and corresponding inference rules for selecting and ranking feature processing processes [72]. He et al. proposed an ontology-based knowledge modeling approach for fast remanufacturing

process planning, which provided a unified framework for managing information and knowledge from different sources [73].

Semantic networks were initially proposed by Quillian in 1968 [74]. It is a structured representation of knowledge of interconnected nodes and arcs that can intuitively represent attributes and semantics. However, it cannot represent dynamic and procedural knowledge. Hao et al. used a priority relation semantic network to reflect the priority relations between processing operations [75]. Zhou et al. used semantic network representation to extract assembly process knowledge from assembly process documents and CAD models, including assembly elements, assembly features, operations, tools, equipment, requirements, and semantic relationships [61]. Zhu et al. proposed an assembly semantic modeling approach for interactive assembly and process generation to improve the efficiency of the assembly process [76]. Gernhardt et al. proposed a semantic process-based knowledge semantic management approach for CAPP in distributed production, which solved the challenge of resource allocation and sharing for CAPP [77]. Shi et al. proposed a semantic knowledge-driven subassembly identification framework to establish a semantic association model between engineering semantic knowledge and assembly models [78]. Morales et al. established semantic networks for collaborative systems based on topology theory and implemented them on the Zeus platform [79].

Framework systems are knowledge representations similar to frame structures that describe knowledge comprehensively. Framework systems are based on human perceptions of what is real and have the weakness of not being flexible enough to express procedural knowledge. Chang et al. represented the process knowledge of rules and instances based on the framework system and then constructed the machining process expert system [80]. Nau and Cheng organized the process knowledge into a specific framework hierarchy and built a SIPP (Semi-Intelligent Process Planner) based on this approach to achieve process planning for metal part manufacturing [81]. Lei et al. proposed a multi-method hybrid framework model for process planning and inference mechanisms. This framework was applied to model the part structure, process decision stages, and the existing set of green process indicators [82]. Ke et al. proposed a disassembly sequence planning method based on frame subgroup structure to improve the disassembly efficiency of electric vehicle batteries [83].

Rule-based knowledge representation is appropriate to represent logical cause–effect knowledge but not to represent structural knowledge. Chen et al. proposed a rule-based process knowledge representation method based on flowcharts [84]. Wang et al. extracted rule knowledge from the complex part machining process to rank different submachine features. A pilot feature-based process planning system was also developed using this approach [85]. Sormaz et al. designed a rule-based intelligent process planning system that was applied to selecting manufacturing processes in metal mechanical parts [86]. Zhou et al. proposed a rule-based process design method to realize the reuse of process knowledge and improve the quality and efficiency of process design [87]. Cakir et al. developed a DieEX expert system that uses IF-THEN rules to represent expertise in planning and machining operations [88].

The graph-based knowledge representation method can represent more complex relationships of things and can avoid some logical constraints. The graph-based knowledge representation includes PG, RDF, and OWL. The models and inference abilities are illustrated in Fig. 4. PG is a directed labeled graph consisting of nodes and relational edges. Both nodes and edges have corresponding properties. The module block of the RDF is the “entity–relationship–entity” triple. RDF is also a directed labeled graph, which provides a logical description of the objective world. OWL is an extension of RDF, which is added to RDF with more semantic components to represent more complex semantic relationships. In the aspect of property graphs, Zhu et al. described the CAD model as the property adjacency graph, where each feature was defined by the property adjacency graph [89]. Ma et al. defined high-level features based on directed acyclic graphs and used

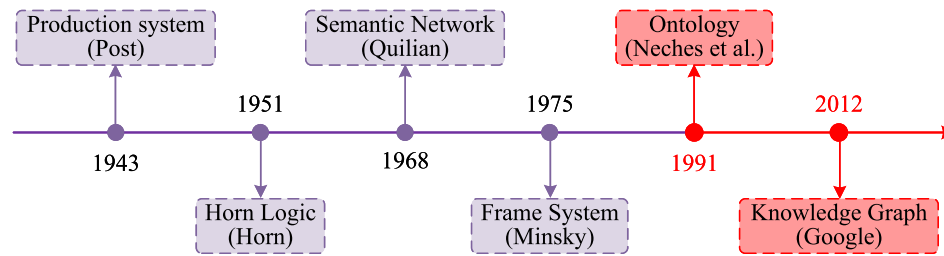


Fig. 3. The milestone techniques of knowledge representation development.

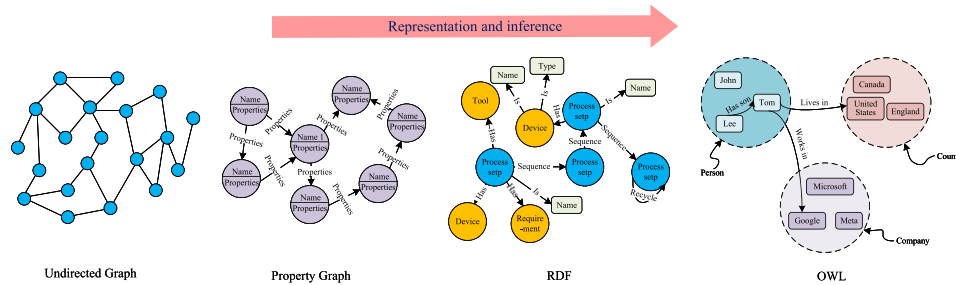


Fig. 4. Structure diagram of the graph-based knowledge representation methods. Knowledge representation and inference capabilities are gradually enhanced from Undirected Graph to OWL.

Table 4
Comparison of knowledge representation methods.

Type	Principles	Features
Rule-based	The rule-based knowledge representation approach is process-centric.	Unstructured knowledge representation; simple inference; simple knowledge systems.
Framework systems	Framework-based knowledge representation is a structured knowledge representation method that resembles the framework storage structure.	Comprehensive knowledge representation; limited flexibility.
Semantic networks	Semantic networks construct network graphs to represent knowledge based on semantic relationships.	Explicitly represent the properties and semantic connections between things; weakly logical.
Ontology-based	The ontology model is a conceptualized and formalized knowledge representation.	Complex knowledge representation capabilities; Internet communication; logical inference; modeling complexity.
PG	The Property Graph is a directed graph, which consists of vertices, edges, attributes, labels, and relational properties.	Simple and flexible structure; no support for deep semantic expressions; no support for logical inference.
RDF	The RDF is a directed graph described by the triple.	Describes relations between entities; simple inference; cannot describe relations and properties between classes; limited representational capabilities.
OWL	The OWL is an extension to RDFs that adds more semantic expression components.	Superior knowledge representation.

a subgraph matching algorithm to identify high-level features from the basic feature priority graph [90]. Naderi et al. studied the integrated process planning and scheduling (IPPS) problem, where each job was represented by a directed network graph [91]. For RDF, Wang et al. used Db2triples to convert the impeller CNC machining instance data into RDF triples and stored them in the neo4j database [92]. OWL has been widely used in representing knowledge of the process to offset the shortcomings of PG and RDF. Sadeghi et al. used OWL ontologies to automate ontology reasoning in process planning for an intelligent process planning system that generates machining instructions for prismatic parts [93]. Qiao et al. proposed a geometrically enhanced ontology web language description logic (OWL-DL) and semantic web rule language (SWRL) inference framework to express assembly sequence planning explicitly [94]. Ye et al. mapped STEP-NC to OWL to describe the knowledge about the machining process of CNC machines [95]. Kim et al. proposed a design for additive manufacturing (DFAM) knowledge

base for additive manufacturing that contains design features, manufacturing features, and parameters. This knowledge base was formalized by OWL and called DFAM ontology [96]. Shi et al. used OWL to define the mapping between spatial objects, assembly processes, and assembly relationships. Furthermore, the feature values and integrated weight values were derived by SWRL rules to construct weighted assembly-directed graphs [78]. Song et al. used OWL to describe the assembly ontology, including class and individual assembly knowledge. The formal description of knowledge about the assembly process was obtained, and the shareability and reusability of assembly knowledge were enhanced [97].

PKG is the rule for describing process knowledge, which is the procedure of symbolization and formalization of knowledge. Each knowledge representation method has strengths and weaknesses. Thus, selecting the PKR method needs to consider the following factors: (1) whether it can adequately represent process knowledge, (2) whether

Table 5
Some statistics of the datasets.

Dataset	Train	Valid	Test	Relations	Triples	Data source
NYT [102]	56196	5000	5000	24	8616	NewYork Times
WebNLG [103]	5019	500	703	216	1607	Wikipedia
WN11 [107]	112581	2609	10544	38696	11	WordNet
WN18 [107]	141442	5000	5000	18	40943	WordNet
WN18RR [108]	86835	3034	3134	11	40943	WordNet
FB13 [109]	316232	5908	23733	13	75043	Freebase
FB15K [107]	483142	50000	59071	1345	14951	Freebase
FB15K-237 [110]	272115	17535	20466	237	14541	Freebase
YAGO3-10 [111]	1079040	5000	5000	37	123182	YOGO

it is conducive to the organization of process knowledge, and (3) whether it is conducive to the utilization and understanding of process knowledge.

5.2. Process knowledge dataset

Process knowledge data are the key assets of manufacturing companies, so public process knowledge datasets are rarely available. The process knowledge dataset focuses on unstructured text data. Although there are very few publicly available process knowledge datasets, many researchers have published their sources of process knowledge or related domain knowledge data. For example, Guan et al. constructed a welding process knowledge dataset from welding manufacturing history process documents [98]. Subsequently, the welding manufacturing process knowledge graph was constructed based on the dataset. Zhou et al. used the processing process documents of an aerospace company as the original data to construct a processing process knowledge dataset and completed to train a process knowledge extraction model on this dataset [99]. In other related fields, such as fault diagnosis [100,101], datasets are constructed on historical fault documents.

Although the related works mentioned above did not make their datasets publicly available, the methods of data generation, data collection, and dataset construction are mostly the same. The methods for constructing these domain knowledge datasets are almost always derived from the NYT [102] and WebNLG [103] datasets. These two datasets are the most well-known open-world public benchmark datasets, and many state-of-the-art knowledge extraction methods are validated on both of them. The NYT dataset was sourced from the New York Times news corpus and produced by the distant supervision method [102]. The WebNLG dataset was sourced from Wikipedia and produced by manual annotation [103]. There are some similar datasets, and Table 5 describes some statistics of the datasets. Inspired by these two benchmark dataset construction methods, domain-specific knowledge datasets have been constructed by researchers. For example, Ling et al. constructed a dataset consisting of 29 528 data text pairs and a vocabulary of 345 words for the weather forecasting domain [104]. Banik et al. constructed a 284 data text pairs dataset for the biology domain [105]. Liu et al. constructed a dataset of 5505 entities and 5512 relations for pharmaceutical descriptions and published the dataset on github [106]. Therefore, inspired by these cases, the process knowledge dataset can be constructed following the method of constructing the NYT/WebNLG dataset. This is also an important direction of our research.

5.3. Construction of process knowledge graph

The PKRs described above are the data organization form of the computer. The PKR is the foundation for constructing the PKG. The construction of PKG includes process knowledge extraction, schema layer construction, and data layer construction. With the process inference module, the complete PKG system is formed. The inference module can generate new process data and feedback on new process knowledge for the PKG. This means that knowledge can be continuously updated. The architecture of PKG is shown in Fig. 5.

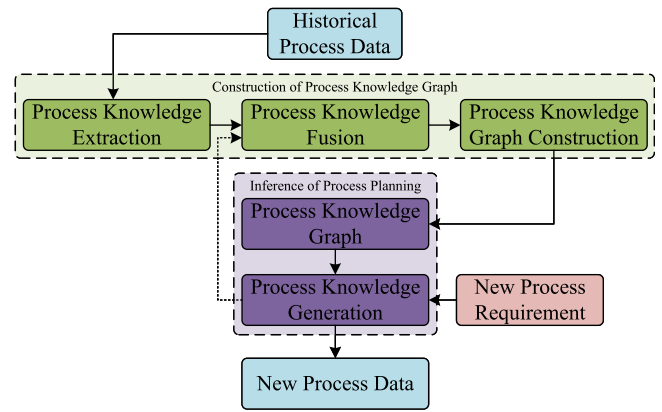


Fig. 5. The flow of PKG construction and application. The dark green part is the flow of PKG construction, and the purple part is the flow of process generation based on PKG. The dotted arrow indicates the knowledge update. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5.3.1. Extraction of process knowledge

Knowledge extraction involves the extraction of knowledge from structured data, semi-structured data, and unstructured data. The workflow and related algorithms for knowledge extraction are shown in Fig. 6. The sources of process knowledge include CAD files and text files. Structured CAD files and unstructured text files are the primary forms of process files. Hence, knowledge extraction of these two file types is the focus of the research.

Extracting process knowledge from the part 3D geometry is indispensable in constructing PKGs. The main methods are feature-based extraction methods and standards for exchange of product model data (STEP) extraction methods (STEP-based extraction methods) [31]. The feature-based extraction methods identify the feature values of the part 3D geometry and use the feature values as inputs of the CAPP system or as process knowledge. In recent decades, numerous research works have been conducted on the part 3D geometric feature extraction for CAPP systems. For example, Liu et al. extracted turned or non-turned feature knowledge of rotating parts from 3D CAD data where the features are concave and convex [112]. Singh et al. simplified the feature extraction process by using a feature library to combine base features, inner features, and boundary features [113]. Joze et al. introduced segmentation methods to divide the 3D model surface region and extracted features as an independent 3D model [114]. Mandfi et al. developed an algorithm for extracting manufacturing features from the process planning procedure, including dimensional and geometric tolerances of machined features [115]. Extracting features from STEP structured files is easier than extracting features from part 3D geometry. The International Organization for Standardization (ISO) defines STEP as a natural data model [116]. STEP is used for swapping data between different applications, such as CAD, CAM, and CAE [117]. STEP files contain geometries, features, structures, tolerances, etc. Sivakumar et al. proposed a feature extraction-based approach to integrate CAD, CAM, and CAI. The generated part model information is extracted from the STEP AP203 file, and this input is used to generate NC code using the developed software [118]. Hallmann et al. assigned manufacturing information, geometric dimensions, and tolerance information to the CAD model from the STEP file on the discrete geometry [119]. Liu et al. extended AP 203 Edition 2 to represent geometric tolerance and dimension information (GD&T) in CAD files in additive manufacturing processes [66]. Zhou et al. used semantic web representation to extract assembly process knowledge from CAD models, including assembly elements and assembly features [61]. Zhu et al. used adjacency graphs to describe CAD models and defined geometric entities and topological information using Boundary representation (B-rep) of solid

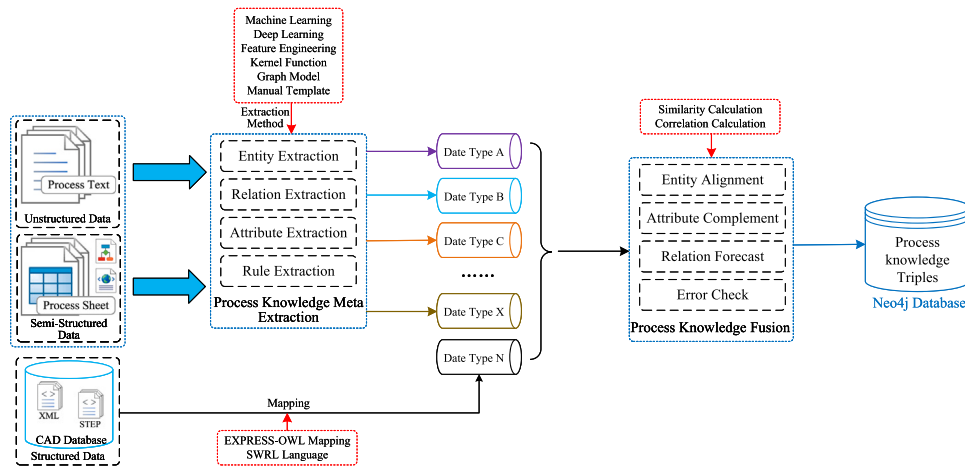


Fig. 6. The flow of process knowledge extraction. The detailed techniques include entity extraction, relation extraction, and attribute extraction. The data source contains structured data, semi-structured data, and unstructured data. The process knowledge triples are stored in graph databases, such as Neo4j.

models [89]. Ghadage et al. proposed a method for generic feature extraction systems for 3D CAD models based on the STEP format [120]. Cui et al. used the internal data of the product CAD model to achieve the extraction of hierarchical information, fit relationships, fit lines, and surface assembly features of the assembly model [121]. Sui et al. converted the STEP information from the output geometric feature information and manufacturing information of the STEP document to the format recognized by the CAPP system [122]. Vyasa et al. used CAD assembly STEP-data as input to extract disassembly-related information to advance remanufacturing and reuse [123].

NLP techniques are used for entity recognition and relation extraction for text process documents. Knowledge extraction from text files supplies the data foundations for PKG construction and process inference. Therefore, how to precisely and efficiently extract knowledge from text files is an important research focus. Gu et al. proposed an assembly semantic entity identification and relationship construction method for assembly process documents, which can effectively obtain assembly process design knowledge from the documents [124]. Zhang et al. proposed a subgraph semantic distance algorithm to extract metallic material entities from DBpedia and construct a metallic material knowledge graph [125]. Kumar et al. proposed a BiLSTM-CRF-based entity recognition method to recognize entities from more than 500,000 manufacturing-related scientific texts, which achieved considerable accuracy [126]. Guo et al. implemented a knowledge extraction framework based on BERT-BiLSTM-CRF for automatic knowledge extraction of process text. This method improved the F-value by 7.35% and 3.87% in the processing domain compared to BiLSTM-CRF and CNN-BiLSTM-CRF, respectively [9]. Guan et al. extracted relationships in unstructured documents based on the BiLSTM+Attention model, and the results showed that high accuracy was obtained in the macro-average metrics [98]. Liu et al. proposed a cognitive matching named entity recognition algorithm to extract entities from large amounts of semi-structured and unstructured data [127].

Process knowledge extraction is a vital technique for constructing PKG. Knowledge extraction from different data sources is the first stage in constructing the PKG. From the published papers, unstructured knowledge extraction is more challenging than the other two. The reasons are that unstructured data sources lack fixed data formats and mapping relationships compared to structured or semi-structured data sources. A large number of unstructured textual process documents are accumulated in industrial production. Therefore, improving the accuracy and speed of unstructured knowledge extraction is an important research direction.

Many state-of-the-art methods have emerged for open-world knowledge extraction, which can also be applied to process knowledge extraction tasks through parameter fine-tuning. The rapid development of

knowledge extraction methods in the open world is due to numerous knowledge datasets [102,103,143–145]. These state-of-the-art knowledge extraction methods have become the foundations for constructing large-scale knowledge graphs, such as DBpedia [49], YAGO [50], Wikidata [51], etc. At present, most knowledge extraction methods use joint extraction to extract entities and relations simultaneously. The technical routes are mainly classified into tagging-based methods, table-filling-based methods, and seq2seq-based methods. The tagging-based methods use binary tag sequences to tag the start and end positions of entities while determining the relations between them [128]. The table-filling-based methods maintain a table for each relation and use table items to denote the start and end positions of entities [131]. The seq2seq-based methods convert the knowledge triple extraction tasks into generation tasks that directly generate entities and relations [136]. For these three technical routes, the current state-of-the-art knowledge triplet extraction methods are summarized and analyzed in Table 6.

5.3.2. Construction of process knowledge graph schema layer and data layer

The PKG is logically divided into the schema layer and the data layer. The schema layer represents the concept classes and relationship definitions of process knowledge, which is the semantic structure of process knowledge. The data layer is the instance representation of the schema layer. The data layer is stored in relational databases such as MySQL, Oracle, SQL Server, etc. According to the concept class and relationship definitions in the schema layer, the data layer consists of entity recognition and relationship construction. The schema and data layers are linked together by data mapping (R2RML language mapping) to form the PKG stored in the graph database. The construction process of PKG is shown in Fig. 7. An example of simple assembly PKG schema layer and data layer is shown in Fig. 8.

With the development of PKG, many researchers have constructed PKGs in the manufacturing field. In 2004, Thimm et al. proposed a graph-theoretic model that represents the design as a data hierarchy tree [146,147]. Later, Wu et al. used a directed assembly connection graph (DACG) to express geometric information and non-geometric knowledge to improve the efficiency of assembly planning [148,149]. The ideology of KG is already contained in these methods. In 2017, Zhao et al. developed an open industrial knowledge graph for manufacturing service matching [15]. In 2018, Li et al. proposed a structured modeling approach for process knowledge mapping for heterogeneous CAM models. A unified modeling framework was established for 3D CAD models and CNC machining processes [150]. In 2020, Bader et al. proposed the Industry 4.0 Knowledge Graph, which describes the latest standards, reference frameworks, and standards [151]. Chen et al.

Table 6

Some state-of-the-art knowledge triples extraction methods. These methods can be migrated to the process knowledge extraction tasks by parameter fine-tuning.

Type	Method	Technical Features
Tagging-based	CasRel [128]	First extract the subject entities, then extract the relations and their corresponding object entities.
	PRGC [129]	First predict the relations, then complete entities extraction and alignment.
	BiRTE [130]	A bidirectional framework for knowledge triples extraction: subject → object → relation and object → subject → relation.
Table filling-based	TPLinker [131]	The joint extraction is viewed as token pair linking and handshaking tagging scheme.
	GRTE [132]	Global features are introduced to model entities and relations and fewer table items.
	UNIRE [133]	Unify the joint extraction of entities and relations into joint labels for filling the tables.
	UniRel [134]	Entity-entity interactions and entity-relation interactions are enhanced in tables.
	OneRel [135]	Similar to TPLinker, but with fewer tables.
Seq2Seq-based	CopyRE [136]	The first model to propose the Seq2Seq method, using a replication mechanism to implement extraction.
	CopyRE+RL [137]	Applying reinforcement learning to the CopyRE model.
	CopyMTL [138]	Multi-task learning capability is enhanced on the CopyRE model.
	SPN [139]	Predicting the triples set with a non-autoregressive parallel decoding approach.
	CGT [140]	A triple extraction model based on generative Transformer.
	REBEL [141]	A Seq2Seq-based relation extraction model that can be applied to longer text data.
	UIE [142]	A unified text-to-structure generative information extraction model that allows for unified modeling for different information extraction tasks.

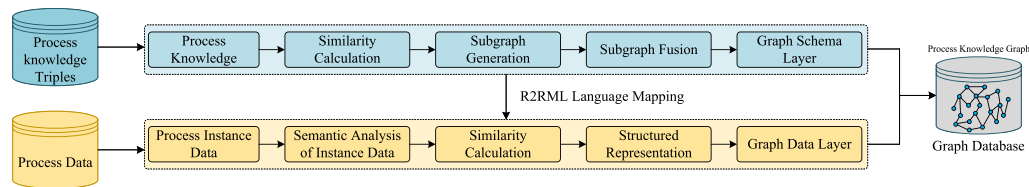


Fig. 7. The construction process of PKG. The upper part shows the construction of the PKG pattern layer, and the lower part shows the construction of the PKG data layer.

constructed the knowledge graph-based assembly information model (KGAM) from the assembly structure pattern layer, the assembly step pattern layer, and the assembly process pattern layer. This layered construction was used to solve the heterogeneous problem of CAD models and assembly process documents [152]. Duan et al. used OWL to establish the pattern layer and constructed the metal cutting process knowledge graph (MCPKG) based on the pattern layer [153]. Subsequently, Guo et al. combined knowledge-driven and data-driven to build a knowledge graph of part information, process knowledge, and equipment resources. The process knowledge was stored in the Neo4j database by instantiating the knowledge graph schema layer [154]. Jiang et al. proposed an assembly PKG construction method based on the digital twin technique. This method provided the possibility for combining the digital twin and the KG in the assembly field [155]. Zhou et al. used an idea similar to hierarchical construction [152] to construct an assembly process knowledge graph (APKG) model [61]. Bachhofner et al. used BPMN2KG (Business Process Model and Notation to Knowledge Graph) [156] to integrate process knowledge, domain knowledge, and dispersed data into an integrated enterprise KG [157]. Wang et al. used Db2triples to convert impeller CNC machining instance data into RDF triples according to pattern layers. A complete KG was formed and stored in the Neo4j database [92]. Dong et al. proposed a process knowledge representation model for heterogeneous models of ships and constructed a multi-element PKG [11].

At present, researchers have constructed many PKGs for specialized domains. For instance, KGAM, MCPKG, APKG, MCPKG, etc. PKGs are more difficult to construct compared to generic KGs. This requires

researchers to master specific process knowledge along with KG construction methods. Moreover, the process data for constructing the PKG is less than the generic KG. This determines that the construction of the PKG is a top-down approach. The schema layer is first constructed and then combined with the process data to complete the mapping to the data layer.

5.4. Refinement of process knowledge graph

The raw PKG constructed directly from the process knowledge triples is incomplete. There may be numerous ambiguous, incomplete, and duplicated knowledge triples in the raw PKG. Thus, it is necessary to deal with these issues by refining the KG. The refinement process includes entity disambiguation, knowledge completion, and knowledge fusion. An illustration is shown in Fig. 9. Although the current refinement methods are for general KGs, these methods can also be used for PKGs.

There are many ambiguous entities named with multiple semantics in historical process files. Entity disambiguation eliminates multiple semantics of entities and maps named entities to unambiguous knowledge. Existing researches focus on local disambiguation and global disambiguation or the combination of both [158]. Local disambiguation deals with a certain range around the entity, which has some noise but with low difficulty [159,160]. Global disambiguation handles the entire graph range to ensure the consistency of entity semantics [161,162]. Some methods combine local disambiguation and global disambiguation [161]. Currently, the techniques used for entity disambiguation

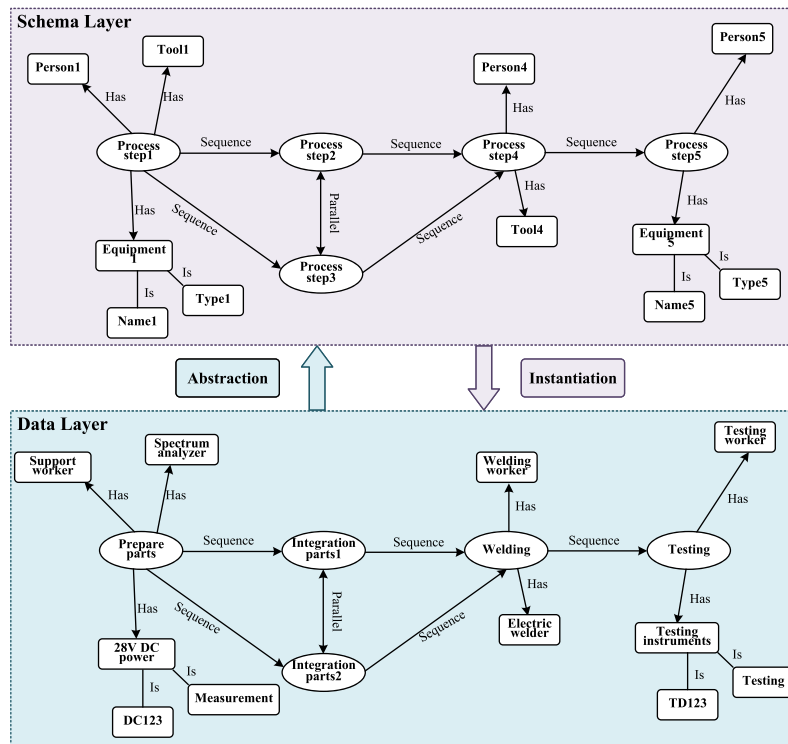


Fig. 8. Instantiation of the process knowledge graph data layer and the schema layer. The upper part is the schema layer, and the lower part is the data layer. From top to bottom is the instantiation of the schema layer, and from bottom to top is the abstraction of the data layer.

are mainly based on deep joint learning and embedding. For local disambiguation, Ganea et al. proposed a novel deep learning model combined with an attention mechanism to handle local entity disambiguation [163]. Fang et al. proposed a joint knowledge and text learning framework for the entity disambiguation task and learned semantically rich embeddings of words and entities [164]. Barba et al. proposed a novel local disambiguation method based on Transformer called ExtEnD [165]. For global entity disambiguation, Yamada et al. proposed a BERT-based global disambiguation model and trained it on the Wikipedia corpus by random mask method [166]. Hu et al. proposed an end-to-end entity disambiguation model GNED that combined local and global information for entity disambiguation [158]. Sevgiliy et al. proposed a graph embedding approach to integrating structured and unstructured text-based information in knowledge graphs, improving entity disambiguation performance [167]. Yamada et al. proposed an embedding-based named entity disambiguation method that jointly maps words and entities into the same continuous vector space [168].

The raw KG is incomplete. Knowledge completion aims to predict and replenish the missing parts of the KG [169]. Knowledge completion fills incomplete knowledge triples and mines new knowledge triples. From the technical aspect, knowledge completion consists of three subtasks: triple classification, link prediction, and relation prediction [170]. Triple classification is to identify the untrue knowledge triples in the KG. For example, Yao et al. proposed a KG-BERT model to implement triple classification and accomplish completion [171]. Jaradeh et al. proposed an exBERT model to implement the triplet classification in academic KGs [172]. Link prediction finds the missing entities based on the existing entities in the KG. Link prediction methods based on distance translation and graphs are widely used, such as TransE [107], TransR [173], TransH [174], SimKGC [175], DisenKGAT [176], etc.

Several process knowledge subgraphs fuse an overall PKG. Furthermore, the newly generated process knowledge also needs to be merged into the PKG. Therefore, knowledge fusion is also essential for refinement [177]. Knowledge fusion includes entity alignment and

attribute alignment. Entity alignment integrates structured data with KG. The representation-based learning methods are the mainstream of entity alignment. Bordes et al. proposed the representation learning model TransE to achieve entity alignment [107]. Chen et al. aligned the entities by linear translation on TransE [178]. The purpose of attribute alignment is to unify attribute notation and semantics. Semantic embedding is the essential method for attribute alignment. For example, Yang et al. used a bag-of-words model to learn contextual semantic embedding of attributes [179]. Sun et al. used the Skip-gram model for attribute embedding [180].

5.5. Validation of process knowledge graph

The validation of the PKG is an essential step to ensure the correctness and accuracy of process knowledge. Currently, validation methods and metrics have been established in generic or open-world KGs [169, 181, 182]. These methods can be used for the validation of PKGs. The validation methods are divided into internal and external methods [183].

External methods use external knowledge to validate the KGs. External knowledge as benchmark datasets are required to have the following characteristics: (1) the external knowledge is highly similar to the validated KG, and (2) the external knowledge is a high-quality benchmark dataset [184]. For example, Li et al. verified the correctness of knowledge graph facts by other knowledge bases and data [185]. Generic KGs describe open-world knowledge that contains highly similar knowledge. Thus, different generic KGs can be cross-validated, i.e., using Freebase to validate DBpedia [186]. For validating PKGs, it is more challenging to use external methods. Because the PKGs are domain-specific KGs, different PKGs describe different process knowledge.

Internal methods use internal data to validate the KGs. The internal data are manually labeled subsets of knowledge triples, and the subsets are used as test sets to validate the KGs [183, 184]. The relevant metrics on the test set can judge the KGs [187]. The internal method is more

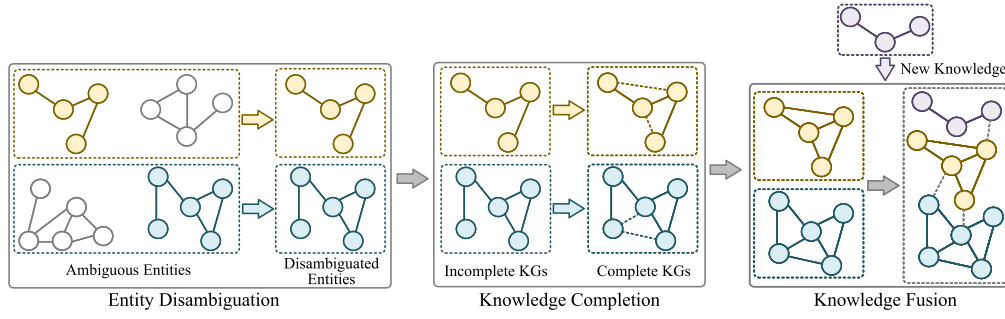


Fig. 9. An illustration of PKG refinement, including entity disambiguation, knowledge completion, and knowledge fusion.

suitable for PKG validation than the external method, although manual annotation of the test set is time-consuming.

The validation metrics [188–190] are as follows:

(1) **Accuracy:** Accuracy is the ratio of knowledge graphs in producing correct predictions, as shown in Eq. (1).

$$Accuracy = \frac{TP + TN}{FN + TP + FP + TN} \quad (1)$$

(2) **Precision:** Precision is the ratio of predicted correct positive samples to predicted positive samples, as shown in Eq. (2).

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

(3) **Recall:** Recall is the ratio of correctly predicted positive samples to all positive samples, as shown in Eq. (3).

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

(4) **F1-score:** F1-score is a trade-off between Precision and Recall, and contains both Precision and Recall, as shown in Eq. (4).

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (4)$$

Where TP is True Positive, FP is False Positives, TN is True Negatives, and FN is False Negatives.

(5) **MRR:** MRR is Mean Reciprocal Rank that calculates the average of inverted triples in the ranking vector. It is used to measure the performance of the knowledge graph in retrieving knowledge triples, as shown in Eq. (5).

$$MRR = \frac{1}{|T|} \sum_{i=1}^{|Q|} \frac{1}{rank(s, p, o)_i} \quad (5)$$

(6) **MR:** MR is Mean Rank that represents the mean of the predicted rankings in the ranking vector, as shown in Eq. (6).

$$MR = \frac{1}{|T|} \sum_{i=1}^{|Q|} rank(s, p, o)_i \quad (6)$$

(7) **Hits@N:** Hits@N denotes the average percentage of knowledge triples with rank $\leq N$ in the link prediction task, as shown in Eq. (7).

$$Hits@N = \begin{cases} \sum_{i=1}^{|Q|} 1 & , rank(s, p, o)_i \leq N \\ 0 & , otherwise \end{cases} \quad (7)$$

Where $rank(s, p, o)_i$ denotes the rank of positive knowledge triples relative to negative knowledge triples, (s, p, o) denotes a knowledge triple, and T is the test set of knowledge triples.

5.6. Knowledge graph-based process generation

The purpose of CAPP is to generate new processes based on production requirements. However, traditional CAPP only simplifies the process design. It cannot adapt to the dynamic changes of the production environment and the variability of process types. Due to its intelligent inference and self-learning, the process knowledge graph can

adapt to the dynamics and variability in industrial production. Process generation based on knowledge graphs has become a popular research direction.

Process inference and generation can be performed on the PKG using inference algorithms. Researchers have combined some inference algorithms with KGs to achieve new processes. For example, retrieval sorting algorithms, matching algorithms, machine learning, deep learning, graph neural networks, etc. A process generation framework based on the similarity matching algorithm is shown in Fig. 10. Pan et al. used a heuristic graph search method to identify each functional sub-assembly in the input model. The automatic planning method for nonlinear assembly sequences was implemented by a graph clustering method to generate hierarchical graphs step by step [191]. He et al. constructed a multi-level manufacturing knowledge graph based on ontology and calculated the similarity between two entities based on semantic similarity [10]. Qian et al. proposed a knowledge and experience-based assembly sequence generation method for automatic assembly sequence inference in Protégé software [192]. Zhou et al. constructed causal event evolution graphs based on KGs and tabular information. Then, the cross-knowledge graphs are fused into a unified knowledge base using a new entity alignment method based on a convolutional network of neighborhood sample graphs [8]. Kesri et al. used NLP techniques to extract an automotive knowledge graph (AutoKG) from process text data. Then, test process cases were generated based on triplet matching and context-based search algorithms [193]. Yi et al. investigated the implementation of hybrid assembly path planning for complex products based on rapidly exploring random trees [194].

For complex product assembly, a multidimensional integrated assembly resource knowledge graph (ARKG) was proposed by Shi et al. Then, it was combined with the Bert model to achieve intelligent planning of product assembly sequences [6]. Liu et al. developed a semantic classification model based on the Naive Bayes Classifier to improve the accuracy of semantic template matching. Furthermore, a knowledge graph-based knowledge Q&A system was established [195]. Duan et al. constructed a metal-cutting process knowledge graph (MCPKG) based on an ontology model. Based on this, the PageRank algorithm was used in conjunction with the data model to recommend tools for process planning [153]. Guo et al. used a hybrid inference algorithm based on semantic analysis and attribute weighting to tackle the problem of heterogeneity in the inference process [154]. The assembly process knowledge graph (APKG) model was developed by Zhou et al. And a sequential knowledge graph convolutional network (SKGCN) model based on distributed graph embedding was designed to generate assembly process planning [61]. Zhang et al. combined KGs with deep learning for macroscopic process planning to improve the efficiency of process generation [196]. Wang et al. used matching algorithms to construct an intelligent Q&A system based on an impeller CNC machining knowledge graph for impeller CNC machining process generation [92]. Dong et al. processed the process knowledge graph based on case-based reasoning (CBR) to retrieve and reuse existing knowledge to generate candidate welding processes [11].

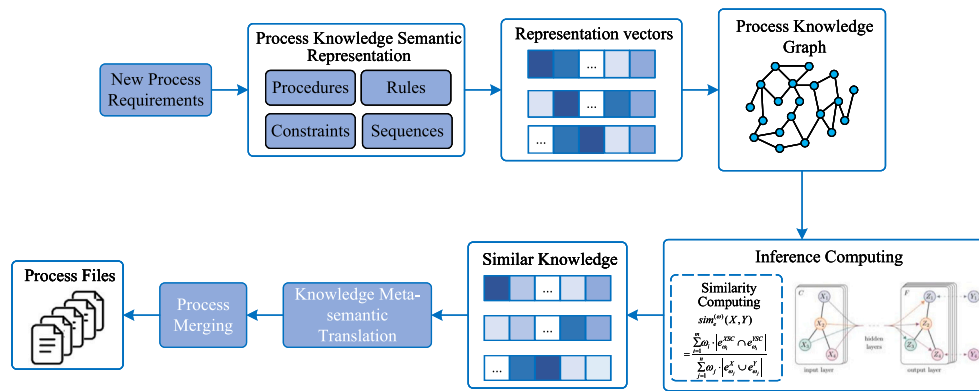


Fig. 10. Framework of the process generation based on similarity matching algorithm.

6. Process knowledge graph combined with large language models

With the development of Generative AI, Large Language Models (LLMs) represented by ChatGPT [197] and GPT4 [198] are creating new booms in artificial intelligence due to their powerful performance. PKGs are structured representations of process knowledge and are extremely interpretable. However, PKGs are difficult to construct in some specific domains, so they are not applied in others. LLMs have powerful generalization capabilities such as extraction, summarization, inference, and generation. However, the LLMs are black-box models that lack factual knowledge and interpretability, which are unacceptable in industrial production. Therefore, PKGs combined with LLMs can take advantage of both. The LLMs can be applied to the construction, refinement, and inference of PKGs.

6.1. PKG construction combined with LLMs

The LLM can be applied in the procedure of PKG construction, such as Named Entity Recognition (NER), Relation Extraction (RE), and Relation Triplet Extraction (RTE). The powerful contextual understanding capability of LLMs is used to extract entities and relations from unstructured historical process text files. There are already related works being published. Li et al. used the LLM to identify entity segments and determine whether they are overlapping or continuous [199]. Yan et al. proposed a Seq2Seq LLM with a pointer mechanism to implement entity extraction [200]. Dai et al. used the BERT large language model to predict contextually relevant entity high-sense words and determine the type of entities [201]. Li et al. inferred entity types using the RoBERTa-large-MNLI model [202]. For relation extraction, Peng et al. used LLM to improve the performance of the relation extraction model [203]. Hong et al. used LLM instead of BiLSTM to achieve document-level relation extraction [204]. Wang et al. utilized LLM to represent global, local, and contextual relations [205].

The development of LLMs may open up new routes for end-to-end KG construction. Guo et al. proposed an end-to-end knowledge extraction model based on BERT [206]. Han et al. proposed a Prompting with Iterative Verification (PiVe) framework that improves the graph-based generation of LLMs [207]. Yasunaga et al. invented a joint training bi-directional model of KG and LLM that can be used for end-to-end kg construction [208]. Kumar et al. proposed an end-to-end text document entity-relation mining method and KG construction method based on an extended pre-trained language model (PLM) [209].

6.2. PKG refinement combined with LLMs

LLMs can be used for PKG refinement, especially for knowledge completion. The encoding and generation capabilities of LLMs can

enhance the ability to process knowledge completion. Then, the knowledge triples are complemented by analyzing the encoded information. Yao et al. utilized a BERT pre-training model to represent knowledge triples as text sequences and encode them [171]. Kim et al. introduced multi-task learning based on the above KG-BERT [171] to improve its effectiveness [210]. Lv et al. transformed knowledge triples into natural language sentences and evaluated their validity using LLMs [211]. Choi et al. introduced a mask entity model in KG completion (MEMKGC) to classify knowledge triples [212]. Wang et al. proposed the SimKGC model, which consists of PLM and contrastive learning framework to improve the performance of knowledge completion [175]. Saxena et al. combined the mask language model with separate coding to enhance knowledge completion in two stages: pre-training and fine-tuning [213]. The common characteristic of the above methods is the LLMs as encoders in knowledge completion.

As generative AI, the Large Language Models have powerful generative capabilities [198,214,215]. Recent researchers have used LLMs as Seq2Seq generators in knowledge completion [216,217], inputting incomplete knowledge triples and outputting complete knowledge triples after completion. Xie et al. proposed a GenKGC to convert the knowledge completion task into the seq2seq generation task. Specifically, entities and relations are used as input sequences to generate the target knowledge triples using a pre-trained LLM [217]. Chen et al. proposed a Seq2Seq model KG-S2S based on the PLM, using the fine-tuned PLM to generate the complete knowledge triples directly [218].

LLMs have their respective advantages as encoders and generators in process knowledge completion. The LLMs as encoders can be easily integrated with previous process knowledge completion methods but require model parameter tuning, which increases the computational workload. The large language models as generators can be applied to process knowledge completion without even tuning the parameters, but the design of prompting in the process planning domain may require some experience. Therefore, there are trade-offs between complexity and computational efficiency in using LLMs to complete process knowledge in specific industrial settings.

6.3. PKG inference combined with LLMs

The LLMs can infer new process routes on the PKGs. Although researches in this field are just emerging, they have broad prospects. There are already some methods in the generic KGs that combine with LLMs for inference. Cao et al. integrated a PLM with an associated KG for joint inference, effectively utilizing the explicit and implicit knowledge contained in the structured KG [219]. Jiang et al. used the ChatCPT for inference on KGs [220]. Zhang et al. implemented facts inference in KGs by exchanging LLMs with the knowledge information in KGs [221].

7. Applications of process knowledge graph

The following summarizes the application of PKG in specialized domains, including assembly, machining, manufacturing safety, and resource allocation.

PKG for Manufacturing Assembly. Assembly process planning is to generate assembly paths for parts in an optimized order. This procedure relies on the knowledge of the process personnel, so the assembly process knowledge lacks efficient management. Moreover, the intelligence of the assembly process evaluation is low, and the optimal assembly sequence cannot be found quickly. As an important tool for knowledge management and inference, KG can handle these problems well. Assembly knowledge is extracted from assembly process documents to construct APKG, and assembly sequence inference is performed on the APKG. The APKG can reduce the dependence on labor and improve efficiency and correctness. Researchers have been constructing domain-specific assembly PKGs. For example, Chen et al. constructed KGAM based on the principle of pattern hierarchy [152]. Zhang et al. proposed AKPG for hierarchical manufacturing [61].

PKGs for CNC Machining. CNC machining process planning includes fixture selection, tool selection, and motion path planning. Each subtask contains many recognition and reasoning tasks, and strong coupling relationships exist between these tasks. According to the complexity of the processing object, the difficulty of CNC machining process planning will be intensified. With the application of the CAPP system in the field of CNC machining, a large amount of data has been accumulated. Knowledge can be extracted from these data to form CNC machining PKG. Duan et al. constructed the MCPKG and enabled to recommend cutting tools for process planning [153]. Wang et al. implemented an intelligent Q&A system for CNC machining based on PKG [92].

PKG for Manufacturing Safety. PKG can also be used in manufacturing processes to detect failures, threats, and process errors to enhance manufacturing safety. Kurniawan et al. used the modular framework of the KG for threat detection, attack graph, and scenario reconstruction [222]. Martinez-Gil et al. proposed a KG for handling faults in the industry based on three action points [223]. Mao et al. proposed a process safety knowledge graph (PSKG) for handling information and risk analysis. A combination of top-down and bottom-up approaches was used to define process safety patterns at the ontology level. Several multi-structured data sources were introduced in building the PSKG. This PSKG enhanced the knowledge-based analysis capabilities and provided the basis for process safety-related applications [224]. Wang et al. developed a new industrial safety knowledge graph (ISKG) and applied it in the coal liquefaction process, which improved the system security [225].

PKG for Production Resource Allocation. PKGs can be used to optimize production resources and achieve optimal resource allocation. Zhou et al. studied the semantic KG generation and reuse method for industrial documents. Based on these techniques, standardized production resources, industry-related knowledge, and Q&A strategies for industrial processing were formed [99]. Zhou et al. also proposed a unified knowledge graph-driven production resource allocation approach for fast resource allocation decisions [13].

8. Future research directions

Through analyzing the development of process planning based on KG in the past decade. The construction of PKG and process generation based on PKG have made rapid advancements. Meanwhile, some difficult issues need to be overcome. Hence, possible future research directions are predicted based on the above overview.

(1) Vector-based PKR

The PKRs reviewed above are based on the discrete symbol approaches. These methods are suitable for representing explicit and discrete process knowledge but inappropriate for representing implicit

process knowledge that cannot be easily symbolized. So, for the semantic completeness of process knowledge, the vector-based approaches need to apply to implicit knowledge [226]. The vectorized representation of knowledge can effectively discover implicit knowledge and potential relationships through numerical computation [107,173]. Thus, we consider vector-based PKR as one of the future research directions and a complement to symbol-based PKR.

(2) Automatic construction of PKG

Currently, PKG depends on manual construction, which is extremely time-consuming. Process knowledge extraction is the key to the automatic construction of PKGs while extracting process knowledge from unstructured texts is a technical difficulty. In recent years, the rapid development of large-scale language pre-training (LLPM) models has allowed automated knowledge extraction [227]. LLPMs have achieved great success in NLP, such as Bert [228], GPT1 [229], GPT2 [230], GPT3 [231], etc. These language processing models can be transferred to process knowledge extraction with fine-tuning or prompting, which will be one of the future research routes to automate the construction of PKGs.

(3) PKG complement

The existing PKGs are relatively static and cannot satisfy the dynamic requirements of production. The dynamic complementary PKG will be the trend of future development to meet the timeliness of process knowledge. Dynamic iterative updating of the PKG can respond to the process requirements of new products. The current complementary approaches based on embeddings [173] and graph convolutional neural networks have been applied to open knowledge graphs [232]. This gives a reference and research route for PKG complement.

(4) Intelligent Q&A-based process generation

The question and answer is the customary way of human communication. Most of the tasks in NLP can be transformed into Q&A forms by linguistic input [233]. This pattern can also be applied to PKG. Intelligent Q&A-based process generation is an upper-level application founded on the PKG, which can make process generation more efficient. The key technology is the combination of PKG with language understanding and semantic search. Firstly, the production requirements are transformed into semantic information that the knowledge graph can understand. New process data is generated from the PKG. Finally, the process data will be transformed into documents that the processors can easily understand. This approach will enhance the usage efficiency of the process in production. So, we regard it as the future direction of research.

9. Discussion and conclusion

Since KGs were formally proposed, using them for process planning has become an important research direction. There are several shortcomings of the traditional CAPP systems compared with the knowledge graph-based process planning:

(1) Traditional CAPP has more manual intervention and often relies on the personal experience of process planners. This leads to too many deciding factors and makes obtaining a measure and an absolute global optimum difficult.

(2) The biggest disadvantage of traditional methods for complex products is that the simulation time can be very long and cannot be applied to large-scale manufacturing of complex products.

(3) The generalization ability of traditional CAPP is relatively weak and cannot be adjusted flexibly.

(4) Fixed planning leads to unbalanced resource loads, lower overall resource utilization, and poor on-time delivery performance.

Based on the above background issues, knowledge graphs can be effective and more intelligent in advancing process planning. It benefits manufacturing enterprises to reduce labor costs, shorten production cycles, and achieve intelligent transformation. This paper summarizes some key technologies and the research status of PKGs in the past ten years. The technical workflows of the knowledge graph-based process planning methods are systematically described. We hope that this review will provide some inspiration to researchers and also contribute to the development of this emerging field.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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